Week 2
Loop-de-Loop
Level 1: Introduction

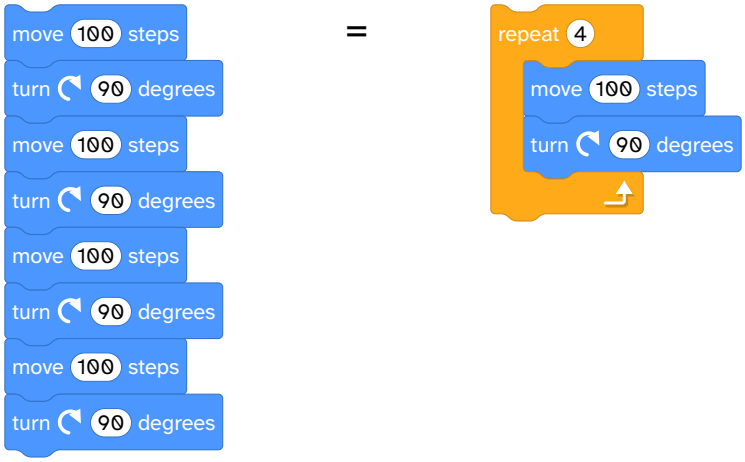
Loops repeat steps so we don't have to.

What you will learn

- How **repeat** and **forever** loops save you from copying blocks.
- How to make a sprite walk by switching **costumes**.
- How to draw with the **Pen** and make shapes with the rule **turn = 360 ÷ sides**.

The big idea

Look at these two scripts. They draw the same square! The left one copies the same two blocks four times. The right one uses a **loop**.

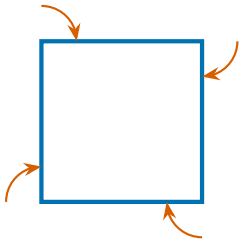


- **repeat 4** runs the blocks inside it 4 times, then the program carries on.
- **forever** runs the blocks inside it again and again until you click the red stop sign.

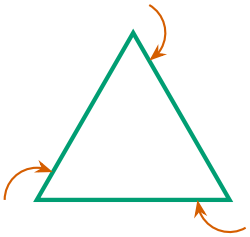
How much to turn?

When you walk all the way around a shape, you turn all the way around once. A whole turn is **360 degrees**. So each corner turn is:

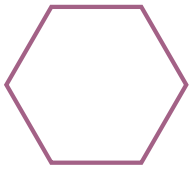
turn = 360 ÷ number of sides



Square: 4 turns
360 ÷ 4 = **90**



Triangle: 3 turns
360 ÷ 3 = **120**



Hexagon: 6 turns
360 ÷ 6 = **60**

The arrows show the turn at each corner. Big arrows = big turns.

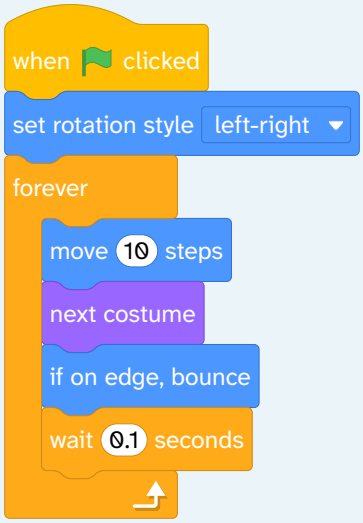
Fill in the table. (Tip: you can let Scratch divide! Put the green "( ) / ( )" block from **Operators** into the turn slot.)

Shape	Sides	Repeat	Turn (degrees)
Triangle	3	3	120
Square	4		
Pentagon	5		
Hexagon	6		
Octagon	8		

Exercise 2.1: Walking Cat

Make the cat walk back and forth forever.

1. Start a new project (**File** → **New**). Click the **Costumes** tab: the cat has 2 costumes. Flipping between them looks like walking! Click **Code** to go back.
2. Build this script. "set rotation style" keeps the cat from walking upside down.
3. Click the flag. Click the red stop sign to stop.

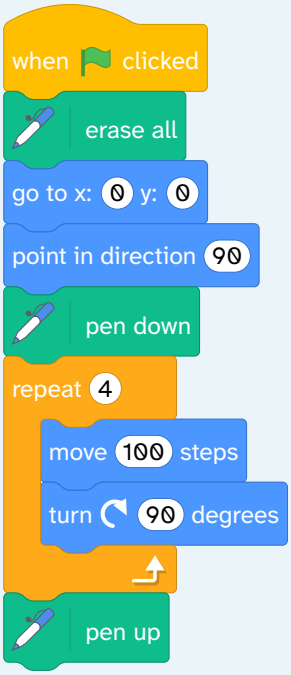


Try it: what happens if you take out the “wait” block?

Exercise 2.2: Polygon Party

Draw shapes with the Pen.

1. Click the blue **Add Extension** button (bottom left) and choose **Pen**.
2. Make the cat small: in the sprite pane, set **Size** to 40.
3. Build the square script. Click the flag. Does the square close?
4. Now change the numbers to draw a triangle, a hexagon and a pentagon. Use your table!



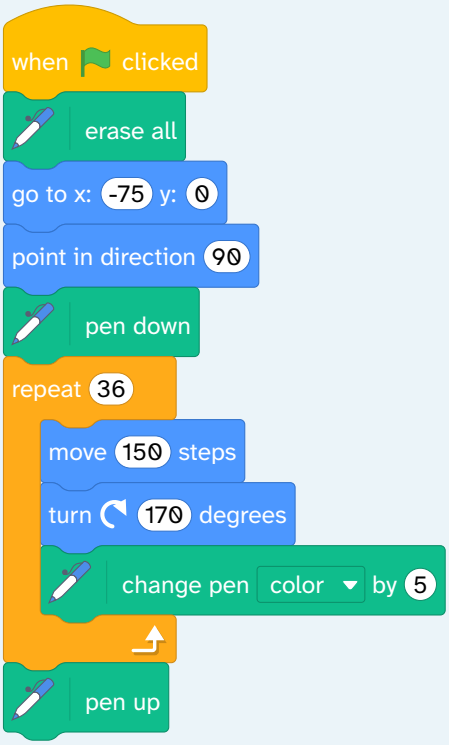
Bug hunt! Leo wanted a triangle. He used repeat **3**, move 100, turn **60**.

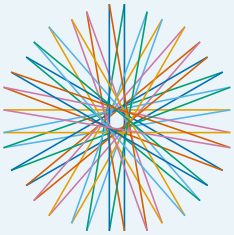
Predict: what will it draw?

Run it. What is the bug, and what is the fix?

Exercise 2.3: Star Burst

Make loop art! Use the same start blocks, then this loop:





What you will see

Now try your own numbers. My favorite: repeat _____ move _____ turn _____

Tip: 20-20-20 for your eyes

Every 20 minutes on a screen, look at something 20 feet (6 meters) away for 20 seconds. Then stretch!

Vocabulary box

loop Code that repeats.

repeat A loop that runs a set number of times, then stops.

forever A loop that keeps going until you click stop.

costume One picture of a sprite. Switching costumes makes animation.

extension Extra blocks you add to Scratch, like the Pen.

degrees How we measure turns. A whole turn is 360 degrees.

Reflection

Where do you see loops in real life?

What was the hardest number to get right today? How did you find it?

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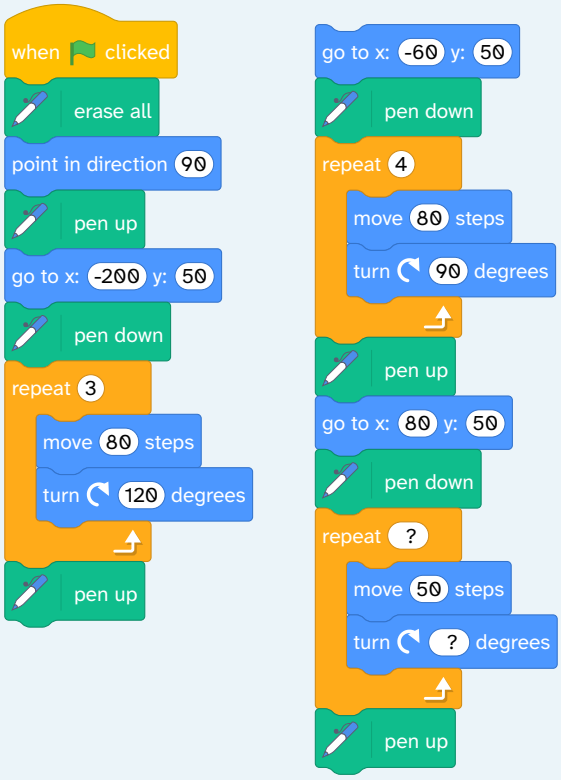
15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Shape Gallery (Scratch, about 10 minutes)

Draw a triangle, a square and a hexagon in a row, like pictures in a gallery.

1. Open Scratch. Click **Add Extension** (bottom left) and choose **Pen**. Set the cat's **Size** to 40.
2. Build the script. "pen up" before each "go to" stops the cat drawing a line between the shapes.
3. Fill in the two missing numbers for the hexagon (use the rule **turn = 360 ÷ sides**).
4. Click the flag. Save as `shape-gallery`.



(The right-hand stack snaps under the left-hand one: it is one long script, split in two to fit the page.)

My hexagon numbers: repeat _____ turn _____

⭐⭐ Challenge (want more?): The circle trick

Draw a shape with “repeat 36”, “move 10 steps”, “turn 10 degrees”. What does it look like? Why do you think it looks like that? (Hint: 36 × 10 = ?)

What turn would you need for a shape with 10 sides? _____

🎨 Creative (make it yours): Your own loop art

Make a picture with loops. Ideas: put “change pen color by 10” inside a loop; put “change pen size by 1” inside a loop; repeat a square 12 times and turn 30 degrees after each one. Draw a tiny sketch of your favorite result here:

✅ Self-check: tick each box when it is true

- ☐ I used “repeat” instead of copying the same blocks again and again.
- ☐ My move and turn blocks are **inside** the repeat.
- ☐ Every shape closes (it ends where it started).
- ☐ I can explain the rule: turn = 360 ÷ number of sides.
- ☐ I did the 20-20-20 eye break when I used the screen.